

# Statistical & Numerical Methods HW2

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*Technion*

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## 1 Area probability with an angle

A needle of length  $l$  is dropped on a grid of parallel lines with distance  $a$  between them, such that  $l < a$ .

1. What is the chance the needle hits the net if it is dropped such that it must be perpendicular to the net?
2. What is the chance it hits the net if it falls at a random angle?

### 1.1 Perpendicular needle

$$1 - \frac{l}{a} \tag{1}$$

### 1.2 at angle

## **2 American Exam**

### **3 Random Coin**

## **4 Random Coin**